

Section 4.1 and 4.2–Continuous Random Variables

Definition (Probability density function (Pdf))

A probability density function or probability distribution function for a continuous random variable X is a function $f(x)$ such that,

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Meaning that probability that random variable X takes any value on the interval $[a, b]$ is the area under the curve $f(x)$ over the interval $[a, b]$.

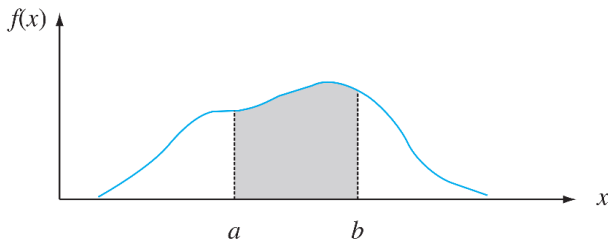


Figure: $P(a \leq X \leq b)$ = the area under the density curve between a and b .

Properties of probability density function $f(x)$ of a continuous random variable X :

(a) $f(x) \geq 0$, for every $x \in \mathbb{R}$

(b) $\int_{-\infty}^{+\infty} f(x)dx = \text{area under the entire graph of } f(x) = 1$

Eg 1. Verify that

$$f(x) = \begin{cases} 12x(1-x)^2 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is a probability density function.

We should show $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx = 1$.

It's clear that $f(x) \geq 0$.

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 12x(1-x)^2 dx = 12 \int_0^1 x[1+x^2-2x] dx$$

$$= 12 \int_0^1 (x + x^3 - 2x^2) dx = 12 \left[\frac{x^2}{2} + \frac{x^4}{4} - \frac{2x^3}{3} \right]_0^1$$

$$= 12 \left[\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right] = 12 \left[\frac{6+3-8}{12} \right] = \frac{12}{12} = 1$$

For a continuous random variable T with pdf $f(t)$, we have

$$\begin{aligned} P(a \leq T \leq b) &= P(a \leq T < b) = P(a < T \leq b) = P(a < T < b) \\ &= \int_a^b f(t) dt \end{aligned}$$

Eg 2. For a particular machine, its useful lifetime (random variable T) is modeled by

$$f(t) = \begin{cases} 2e^{-ct} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find c such that f is a valid probability density function.

Find the probability that the machine will last

(b) less than 2 years

(c) more than 2 but less than 4 years

(d) more than 4 years

$$(a) \quad \int_0^{\infty} 2e^{-ct} dt = 1 \Rightarrow \lim_{a \rightarrow \infty} \int_0^a 2e^{-ct} dt = 1$$

Let us first calculate $\int 2e^{-ct} dt = ?$

We can use u substitution $u = -ct \Rightarrow du = -c dt \Rightarrow -\frac{du}{c} = dt$

$$2 \int e^{-ct} dt = \frac{2}{-c} \int e^u du = -\frac{2}{c} e^u = -\frac{2}{c} e^{-ct}$$

$$\lim_{a \rightarrow \infty} \int_0^a 2e^{-ct} dt = 1 \quad \Rightarrow \quad \lim_{a \rightarrow \infty} \left. -\frac{2}{c} e^{-ct} \right|_0^a = 1$$

$$-\frac{2}{c} \lim_{a \rightarrow \infty} e^{-ca} + \frac{2}{c} e^0 = 1 \Rightarrow -\frac{2}{c} \lim_{a \rightarrow \infty} e^{-ca} + \frac{2}{c} = 1 \quad (*)$$

If $c < 0$ then, $\lim_{a \rightarrow \infty} e^{-ca} = \infty$, which $(*)$

will be impossible. Hence $c > 0$, $\lim_{a \rightarrow \infty} e^{-ca} = 0$

$$\Rightarrow \frac{2}{c} = 1 \Rightarrow \boxed{2 = c}$$

$$\begin{aligned}
 \text{(b)} \quad P(T < 2) &= P(T \leq 2) = \int_{-\infty}^2 f(t) dt = \int_0^2 2e^{-2t} dt \\
 &= -e^{-2t} \Big|_0^2 = -e^{-4} + 1 = 1 - e^{-4}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad P(2 < T < 4) &= \int_2^4 f(t) dt = \int_2^4 2e^{-2t} dt = -e^{-2t} \Big|_2^4 \\
 &= e^{-4} - e^{-8}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad P(T > 4) &= \int_4^{\infty} 2e^{-2t} dt = \lim_{a \rightarrow \infty} \int_4^a 2e^{-2t} dt = \lim_{a \rightarrow \infty} (-e^{-2t}) \Big|_4^a \\
 &= -\lim_{a \rightarrow \infty} e^{-2a} + e^{-8} = e^{-8}
 \end{aligned}$$

Definition (Cumulative Distribution function (CDF))

For a continuous random variable X with pdf $f(x)$ we define cumulative distribution function CDF $F(x)$ as follows

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y)dy$$

Note: By Fundamental Law of Calculus, the above definition tells that derivative of the CDF of a continuous random variable is its pdf. That is

$$\frac{dF}{dx}(a) = F'(a) = f(a)$$

Note: CDF $F(x)$ a continuous random variable X is always continuous.

Note: $0 \leq F(x) \leq 1$

Now we can complete the previous slide

For a continuous random variable T with pdf $f(t)$ and cdf $F(t)$, we have

$$\begin{aligned} P(a \leq T \leq b) &= P(a \leq T < b) = P(a < T \leq b) = P(a < T < b) \\ &= \int_a^b f(t) dt = F(b) - F(a) \end{aligned}$$

Eg 3. Given the probability density function

$$f(x) = \begin{cases} 6x(1-x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find cumulative density function $F(x)$.

(b) $P(X \leq 1)$

(c) $P(X \geq 3)$.

(d) $P(\frac{1}{3} \leq X < \frac{1}{2})$.

$$F(x) = \begin{cases} 0 & x < 0 \\ 3x^2 - 2x^3 & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

if $0 \leq a \leq 1$

$$\begin{aligned} F(a) &= \int_0^a 6x(1-x) dx \\ &= 6 \int_0^a (x - x^2) dx \\ &= 3x^2 - 2x^3 \Big|_0^a \\ &= 3a^2 - 2a^3 \end{aligned}$$

$$(b) P(X \leq 1) = F(1) = 1$$

$$(c) P(X \geq 3) = 1 - P(X < 3) = 1 - P(X \leq 3) \\ = 1 - F(3) = 1 - 1 = 0$$

$$(d) P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right) = F\left(\frac{1}{2}\right) - F\left(\frac{1}{3}\right) \\ = \left[3\left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right)^3\right] - \left[3\left(\frac{1}{3}\right)^2 - 2\left(\frac{1}{3}\right)^3\right] \\ = \left[\frac{3}{4} - \frac{1}{4}\right] - \left[\frac{3}{9} - \frac{2}{27}\right] = \frac{1}{2} - \frac{7}{27} = \frac{27-14}{54} = \frac{13}{54}$$

Eg 4. A continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x^3}{27} & 0 \leq x \leq 3 \\ 1 & x > 3 \end{cases}$$

(a) Find its probability density function $f(x)$.

(b) Find point x such that 25% of data points are below this point. (or find the 1st quartile Q_1)

$$f(x) = \begin{cases} F'(x) = \frac{3}{27} x^2 = \frac{x^2}{9} & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \frac{x^2}{9} & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

(b) Find point a such that $P(X \leq a) = \frac{1}{4}$.

$$\Rightarrow \text{Find } a \text{ s.t. } F(a) = \frac{1}{4}$$

$$\frac{1}{4} = \frac{a^3}{27} \Rightarrow a^3 = \frac{27}{4}$$

$$\Rightarrow a = \sqrt[3]{\frac{27}{4}} = \frac{3}{\sqrt[3]{4}}$$